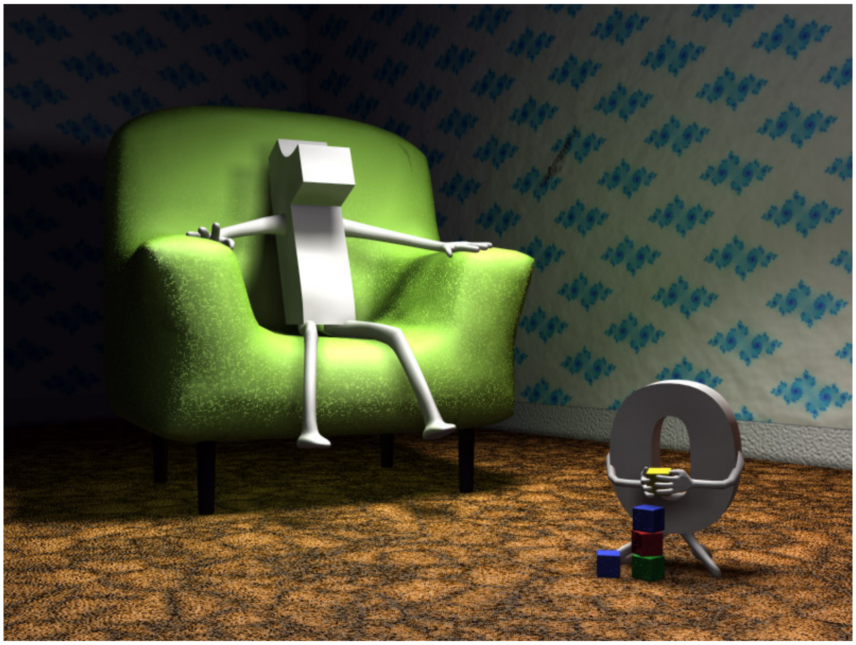
Chapter 1. INTEGERS

1. 3 Integers modulo n

1.3.1 Modular arithmetic



Today we will introduce an addition and a multiplication on Z_n, and see some of their properties.

Addition and multiplication modulo n

for $[a], [b] \in \mathbb{Z}_n$ we define:

$$[a] + [b] = [a + b] \quad \text{and} \quad [a] \cdot [b] = [ab].$$

Examples: $[3]_6 + [5]_6 = [3+5]_6 = [8]_6 = [2]_6$

$\mathbb{Z}_6 = \{[0]_6, [1], \ldots, [5]_6\}$ because $8 \equiv 2 \pmod 6$

$[3]_6 \cdot [5]_6 = [3 \cdot 5]_6 = [15]_6 = [3]_6$ because $15 \equiv 3 \pmod 6$

Theorem 1.38. *Let $n \geq 2$ be an integer, and $a_i, b_i \in \mathbb{Z}$ (for $i = 1, 2$) such that $a_1 \equiv a_2 \pmod n$ and $b_1 \equiv b_2 \pmod n$. Then:*

(1) $a_1 + b_1 \equiv a_2 + b_2 \pmod n$,

(2) $a_1 b_1 \equiv a_2 b_2 \pmod n$,

(3) $a_1^m \equiv a_2^m \pmod n$, *for all $m \geq 1$.*

$$\begin{array}{ccccc}[a_1]_n & + & [b_1]_n & = & [a_1+b_1]_n \\ \| & & \| & & \|? \\ {[a_2]_n} & + & [b_2]_n & = & [a_2+b_2]_n\end{array}$$

<u>Proof</u> $a_1 \equiv a_2 \pmod{n} \Rightarrow n \mid (a_1 - a_2)$

$b_1 \equiv b_2 \pmod{n} \Rightarrow n \mid (b_1 - b_2)$

<u>(1)</u> By thm 1.10 (4) we have that

$n \mid \left((a_1 - a_2) + (b_1 - b_2) \right)$, that is,

$n \mid \left((a_1 + b_1) - (a_2 + b_2) \right)$, which implies that

$a_1 + b_1 \equiv a_2 + b_2 \pmod{n}$.

(2) Applying thm 1.10 (4)

$n \mid (a_1 - a_2) b_1$ and $n \mid (b_1 - b_2) a_2 \Rightarrow$

$\Rightarrow n \mid \left((a_1 - a_2) b_1 + (b_1 - b_2) a_2 \right) \Rightarrow n \mid (a_1 b_1 - a_2 b_2)$

$\Rightarrow a_1 b_1 \equiv a_2 b_2 \pmod{n}$ $\blacksquare$

Notation 1.39. From now on, to simplify our calculations, we will denote the residue class $[a]_n$ simply by $\bar{a}$.

Properties of the modular operations

Theorem 1.42. *Let $n \geq 2$ be a fixed modulus and $a, b, c \in \mathbb{Z}$. Then the following hold in $\mathbb{Z}_n$.*

(1) $\bar{a} + \bar{b} = \bar{b} + \bar{a}$ *and* $\bar{a}\bar{b} = \bar{b}\bar{a}$.

(2) $\bar{a} + (\bar{b} + \bar{c}) = (\bar{a} + \bar{b}) + \bar{c}$ *and* $\bar{a}(\bar{b}\bar{c}) = (\bar{a}\bar{b})\bar{c}$.

(3) $\bar{a} + \bar{0} = \bar{a}$ *and* $\bar{a}\bar{1} = \bar{a}$. We call $\bar{0}$ the **zero** of $\mathbb{Z}_n$, while $\bar{1}$ is said to be the **unity** of $\mathbb{Z}_n$.

(4) $\bar{a} + \overline{-a} = \bar{0}$. *We usually write $-\bar{a}$ to denote $\overline{-a}$.*

(5) $\bar{a}(\bar{b} + \bar{c}) = \bar{a}\bar{b} + \bar{a}\bar{c}$.

Remark 1.43. There are **several differences** between the **arithmetic** of $\mathbb{Z}_n$ and that of $\mathbb{Z}$. For example:

- If $ab = 0$ in $\mathbb{Z}$, then either $a = 0$ or $b = 0$. This need not hold in $\mathbb{Z}_n$. For instance, in $\mathbb{Z}_6$ we have that $\bar{2} \cdot \bar{3} = \bar{0}$, and $\bar{2} \neq \bar{0}$, $\bar{3} \neq \bar{0}$.

As a consequence in $\mathbb{Z}$ we know that if $a \neq 0$ and $ab = ac$, then $b = c$. This is no longer true!

$\underline{\mathbb{Z}_6}$ $\quad \bar{2} \cdot \bar{3} = \overline{2 \cdot 3} = \bar{6} = \bar{0}$ on $\mathbb{Z}_6$.

and $\bar{2} \neq \bar{0}$ and $\bar{3} \neq \bar{0}$ in $\mathbb{Z}_6$.

the cancellation law might not hold in $\mathbb{Z}_n$.

$$\bar{4} \cdot \bar{2} = \overline{4 \cdot 2} = \bar{8} = \bar{2} \text{ in } \mathbb{Z}_6$$

$$\|$$

$$\bar{4} \cdot \bar{5} = \overline{4 \cdot 5} = \overline{20} = \bar{2} \text{ in } \mathbb{Z}_6$$

But $\bar{2} \neq \bar{5}$ in $\mathbb{Z}_6$.

$\boxtimes$